Lotka-Volterra Systems: Dynamics and Applications

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Predator – prey models

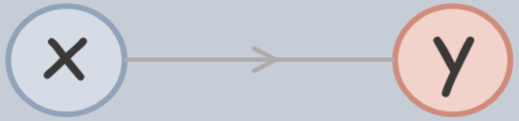
$$\frac{dx}{dt} = x - xy$$
$$\frac{dy}{dt} = y + xy$$

Ecology

Epidemics

Finance

Mathematical Modelling: A Predator-Prey System



The Lotka-Volterra model describes the evolution of two species, where one represents the predator and the other the prey. Their populations change in time according to the two first-order *nonlinear* differential equations:

$$\begin{aligned}\frac{dx}{dt} &= \alpha x - \beta xy \\ \frac{dy}{dt} &= -\gamma y + \delta xy\end{aligned}$$

where $\alpha, \beta, \gamma, \delta > 0$.

Key-Features

- The Lotka-Volterra model is a dynamical system which was introduced independently by Lotka and Volterra as a predator-prey model.

- A predator-prey model is widely used to describe the dynamical evolution of systems in ecology, biology and finance.

- It is a toy model: Not very realistic, though it works rather well for large enough data sets. Contains the minimum possible number of variables and assumptions.

A.J. Lotka (1880–1949) was a bio-mathematician and a bio-statistician

✧ applied principles of the physical sciences to biological sciences

✧ Lotka's principle: the competition among organisms for available energy is a natural selection.

[1] A. J. Lotka (1939) *Analytical theory of biological populations.*

V. Volterra (1860–1940) was a mathematician and physicist.

✧ strongly influenced the modern development of calculus.

✧ drew the attention of mathematicians to mathematical biology.

[2] V. Volterra (1931) *Leçons sur la théorie mathématique de la lutte pour la vie.*

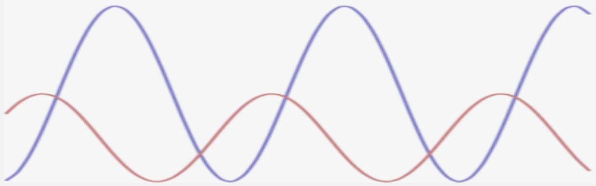
Mathematical Modelling:
A Predator-Prey System

1. Choose the system and its **variables**

2. Describe system’s **dynamics**

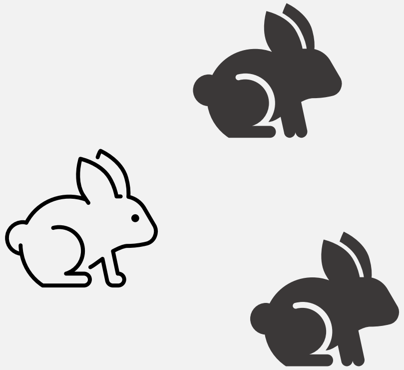
$$\frac{dx}{dt} \qquad \frac{dy}{dt}$$

3. **Solve** the system

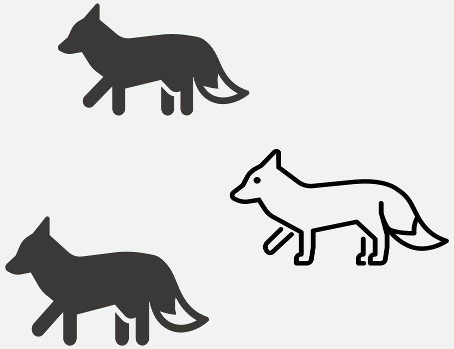


Mathematical Modelling: A Predator-Prey System

1\. Choose the system and its **variables**



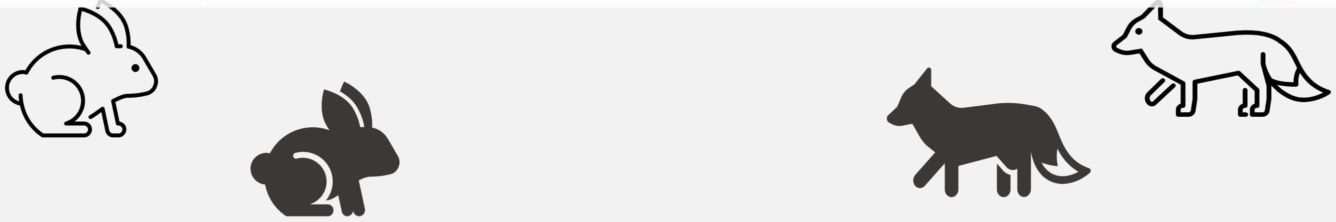
rabbits



foxes

Mathematical Modelling: A Predator-Prey System

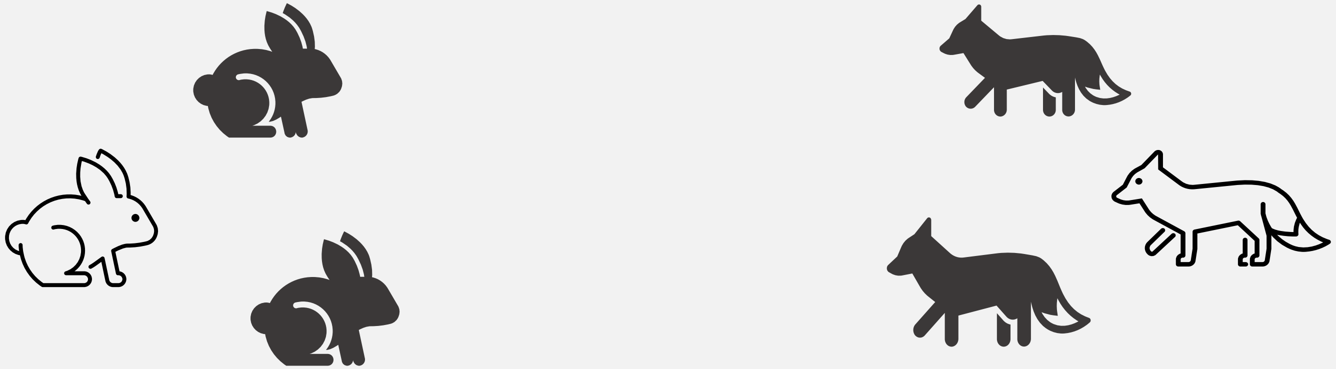
- x is the population of prey (e.g. rabbits)
- y is the population of predators (e.g. foxes)



rabbits *foxes*

Mathematical Modelling:
A Predator-Prey System

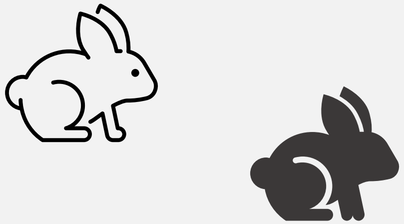
2. Describe system's **dynamics**

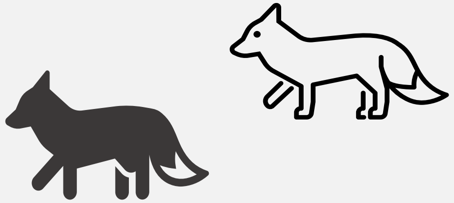


Mathematical Modelling: A Predator-Prey System

2. Describe system's **dynamics**

- dx/dt is the population growth rate of prey
- dy/dt is the population growth rate of predators





Mathematical Modelling: A Predator-Prey System

3. **Solve** the system

$$\frac{dx}{dt} = \alpha x - \beta xy$$
$$\frac{dy}{dt} = -\gamma y + \delta xy$$

where $\alpha, \beta, \gamma, \delta > 0.$

Dynamical Systems Theory

2.8 The Hartman–Grobman Theorem

The Hartman–Grobman Theorem is another very important result in the local qualitative theory of ordinary differential equations. The theorem shows that near a hyperbolic equilibrium point $\mathbf{x}_0$, the nonlinear system

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}) \tag{1}$$

has the same qualitative structure as the linear system

$$\dot{\mathbf{x}} = A\mathbf{x} \tag{2}$$

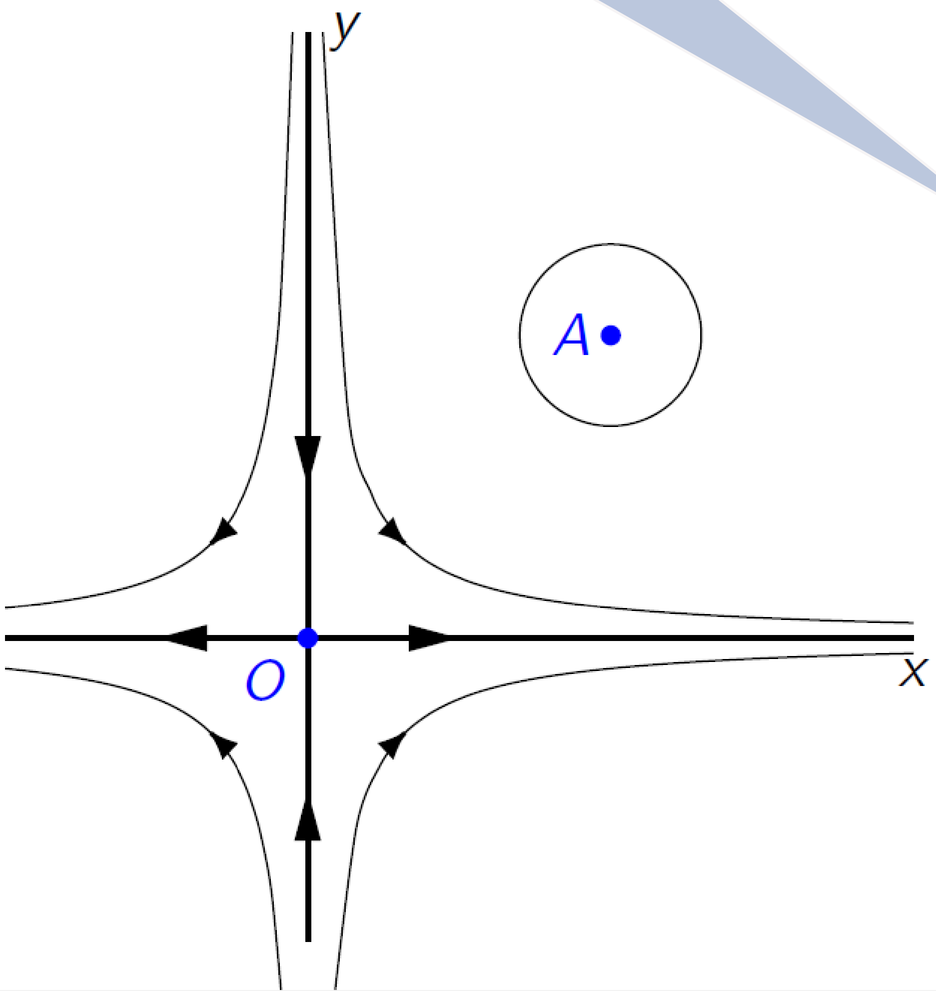
with $A = D\mathbf{f}(\mathbf{x}_0)$.

Definition 1. Two autonomous systems of differential equations such as (1) and (2) are said to be *topologically equivalent* in a neighborhood of the origin or to have the *same qualitative structure near the origin*

Reference:
L. Perko
Differential Equations and Dynamical Systems,
Springer
(2000)

The Lotka-Volterra Phase Portrait

It's all about stability!



- At $O(0,0)$ it is:

$$Df(0,0) = \begin{pmatrix} \alpha & 0 \\ 0 & -\gamma \end{pmatrix}$$

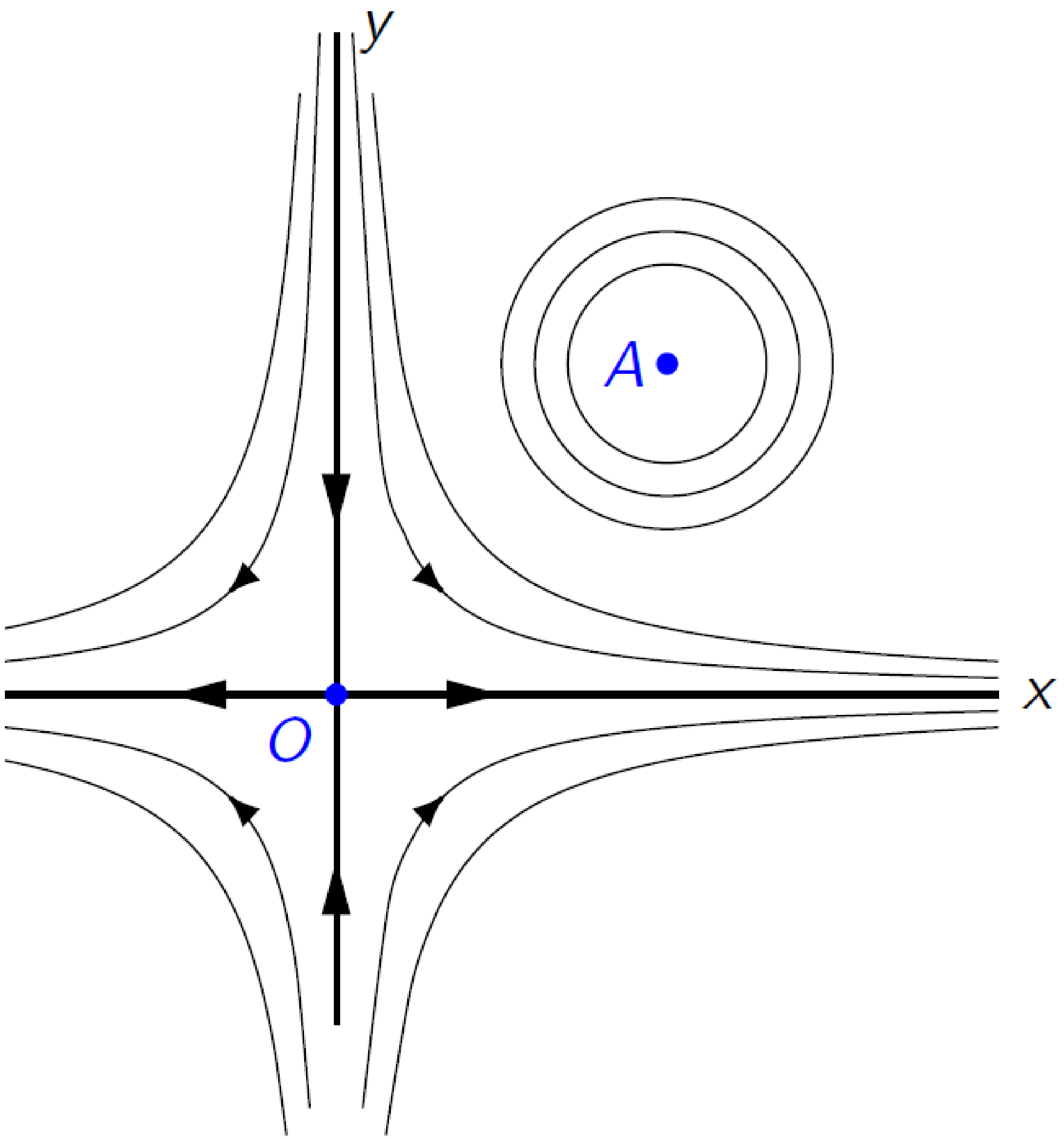
so has two real eigenvalues $\lambda_1 = \alpha$ and $\lambda_2 = -\gamma$ therefore $O(0,0)$ is a **saddle**.

- At $A(\frac{\gamma}{\delta}, \frac{\alpha}{\beta})$ it is:

$$Df(\frac{\gamma}{\delta}, \frac{\alpha}{\beta}) = \begin{pmatrix} 0 & -\beta\frac{\gamma}{\delta} \\ \delta\frac{\alpha}{\beta} & 0 \end{pmatrix}$$

and has the pair of pure imaginary complex conjugate eigenvalues $\lambda_{1,2} = \pm \mathrm{i}\sqrt{\alpha\gamma}$, therefore $A(\frac{\gamma}{\delta}, \frac{\alpha}{\beta})$ is a **center**.

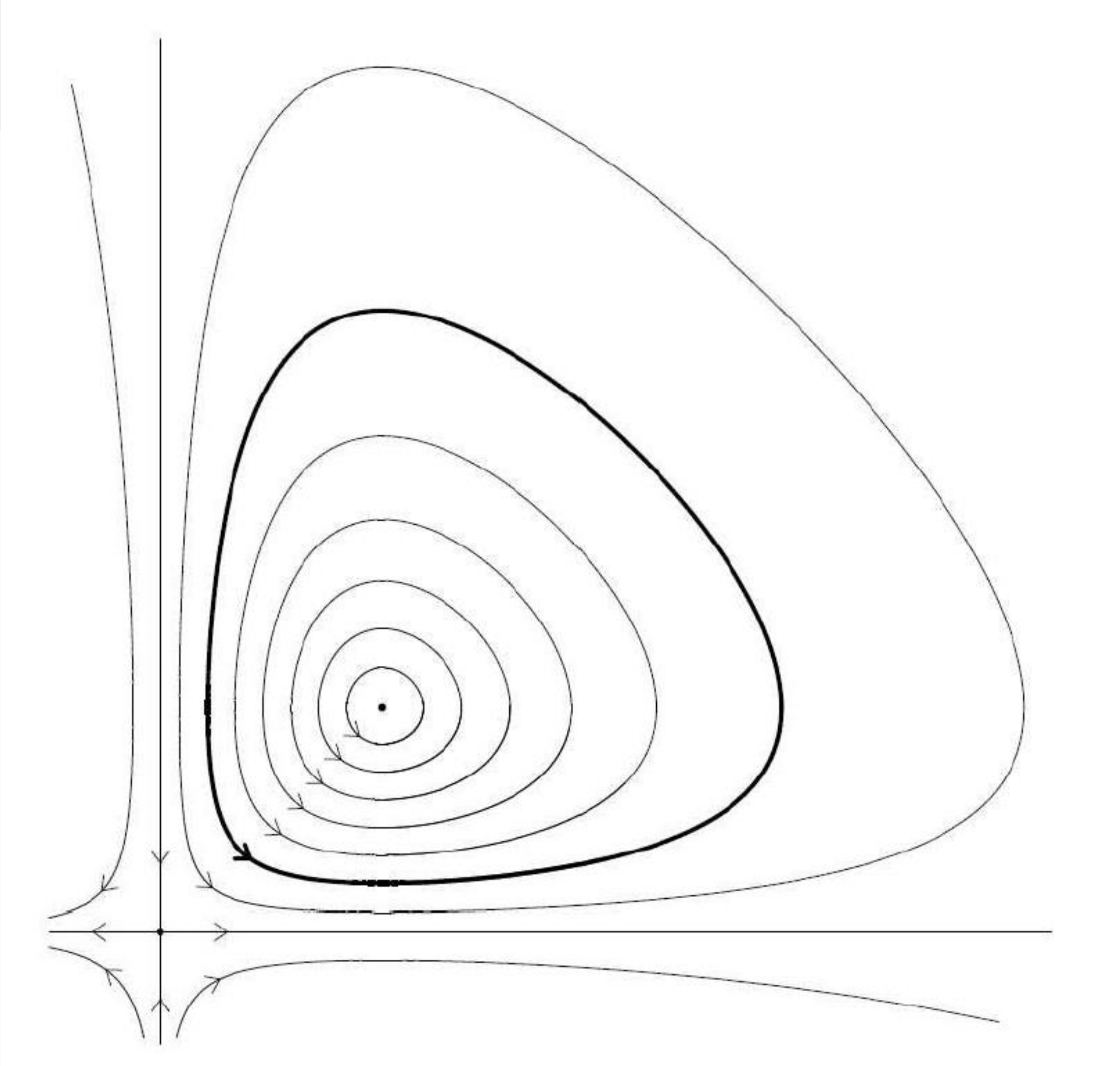
The Lotka-Volterra Phase Portrait



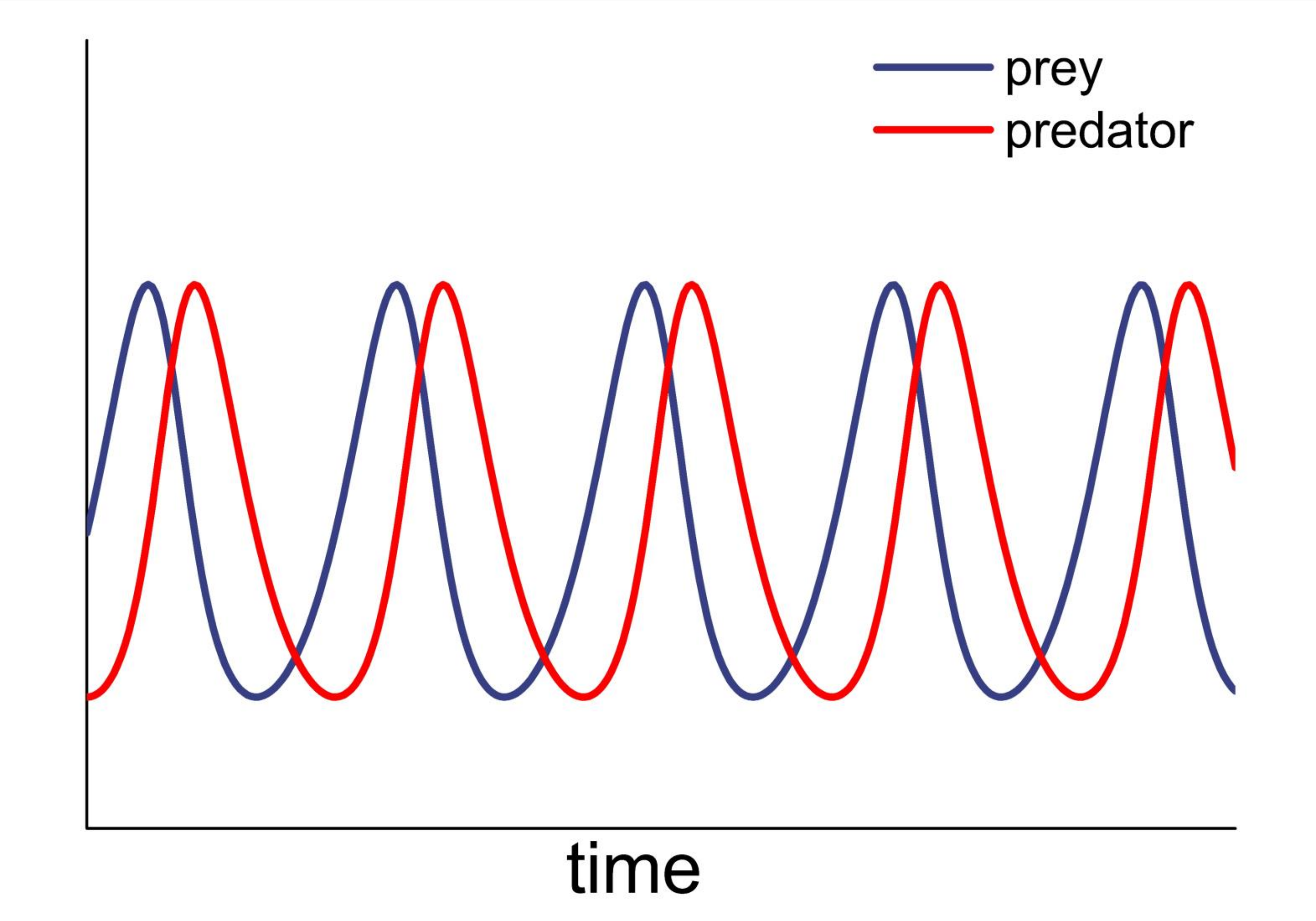
The orbits around both equilibrium points should be compatible.

The direction of the flow should also be compatible.

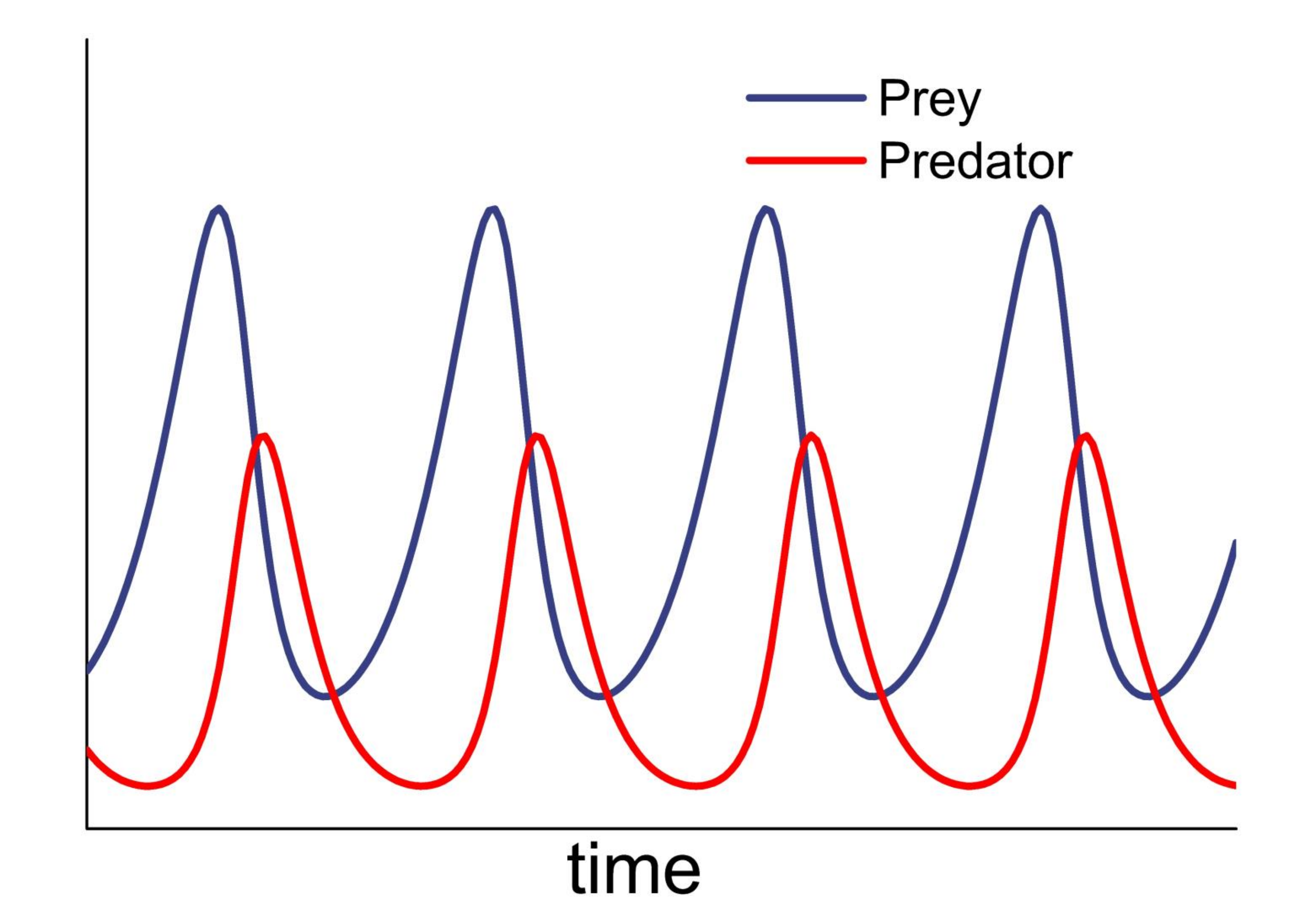
The Lotka-Volterra Phase Portrait



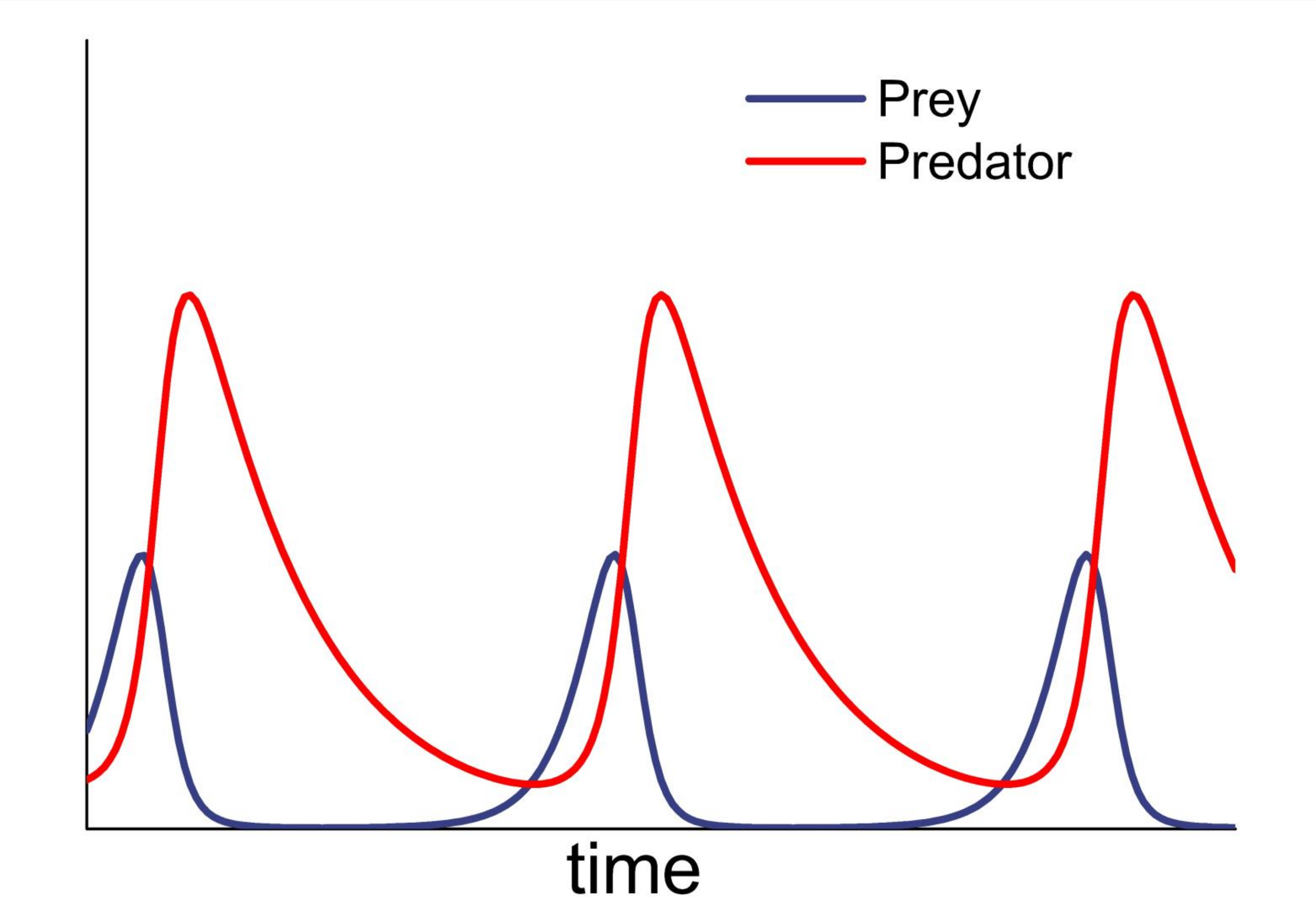
Periodic Solutions



Periodic Solutions



Periodic Solutions



Application in Epidemics

$$\frac{dx}{dt} = -\varepsilon\, xy$$

$$\frac{dy}{dt} = -y + \varepsilon\, xy$$

$\varepsilon > 0$ very small

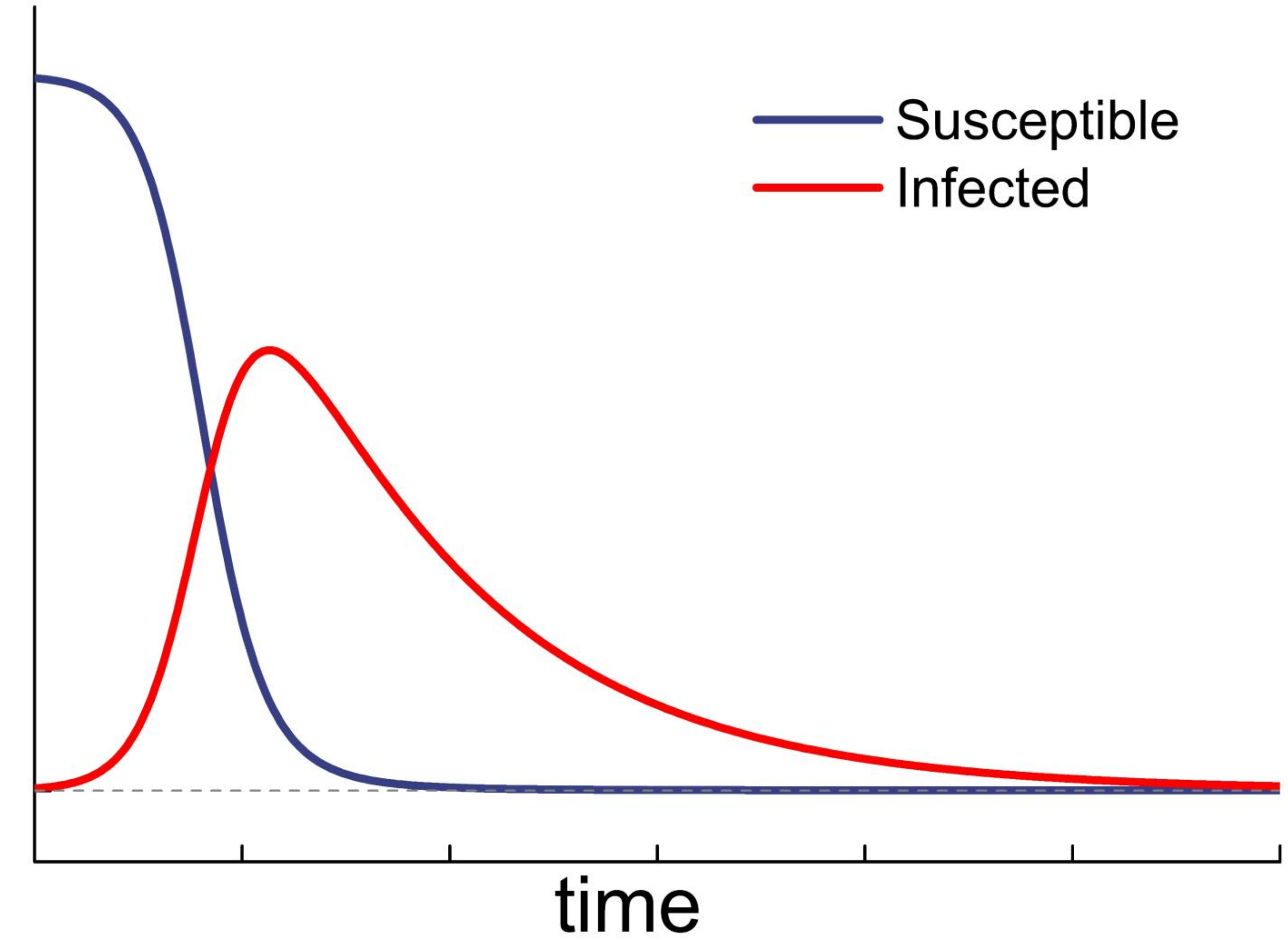
Epidemic Model

Application in Epidemics

$$\frac{dx}{dt} = \qquad -\varepsilon\, xy$$

$$\frac{dy}{dt} = -y + \varepsilon\, xy$$

$\varepsilon > 0$ very small

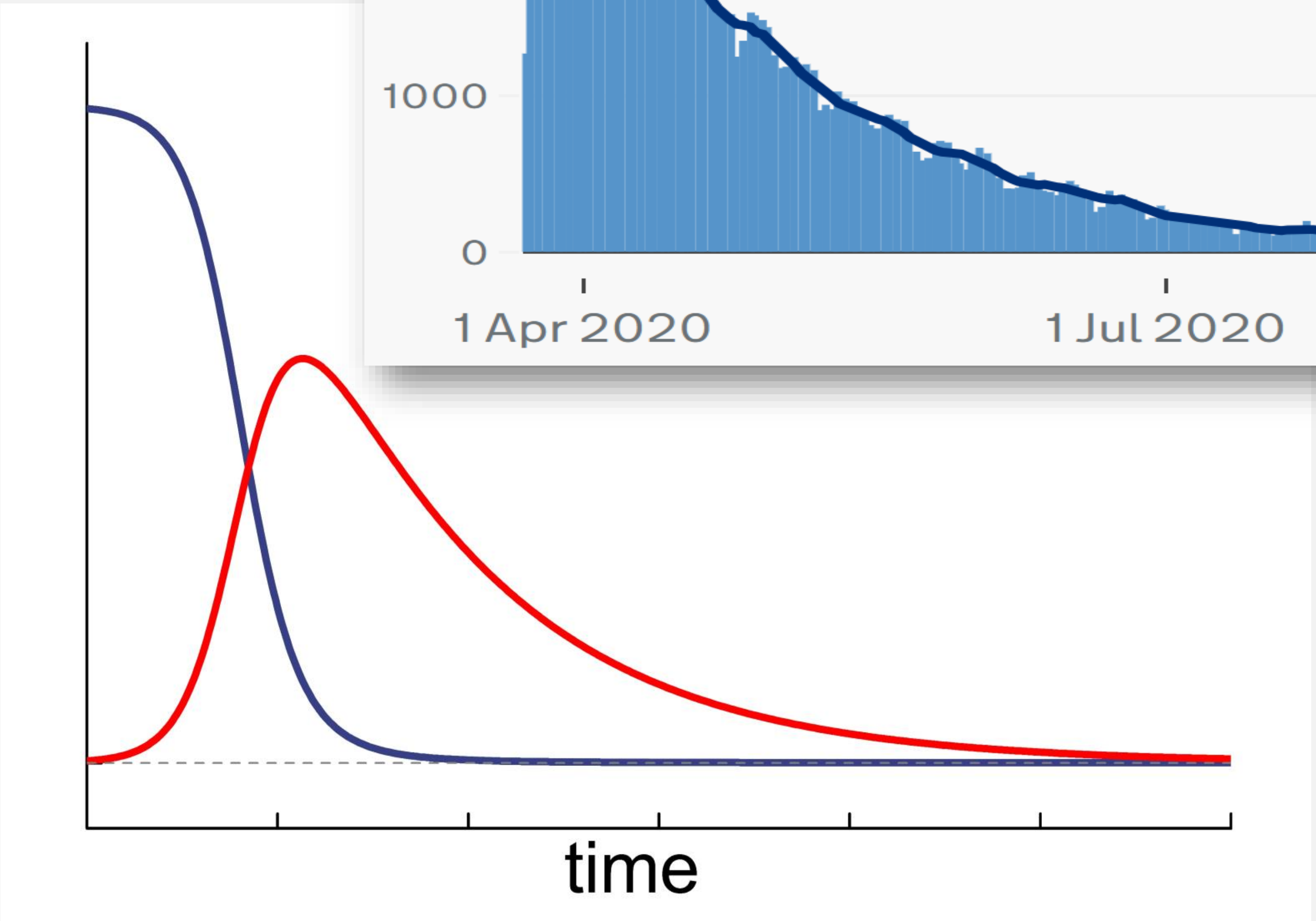


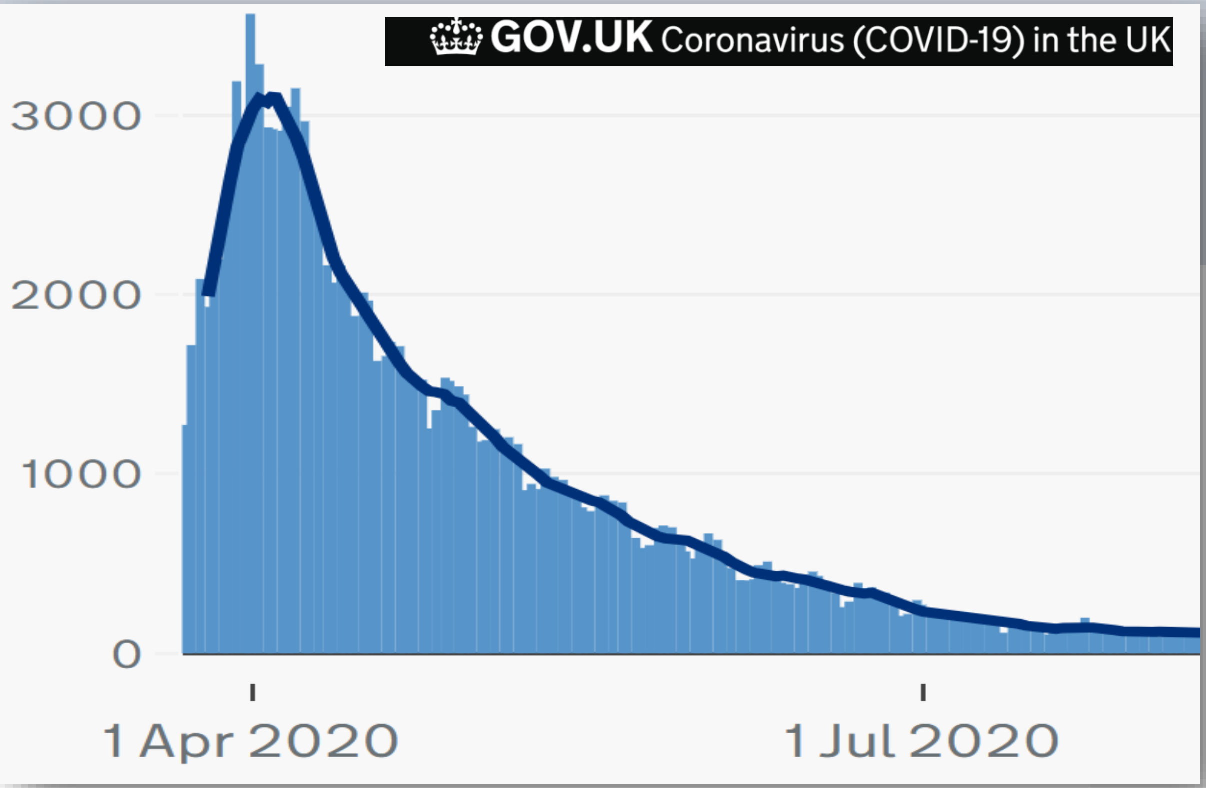
Application in

$$\frac{dx}{dt} = \qquad -\varepsilon\, xy$$

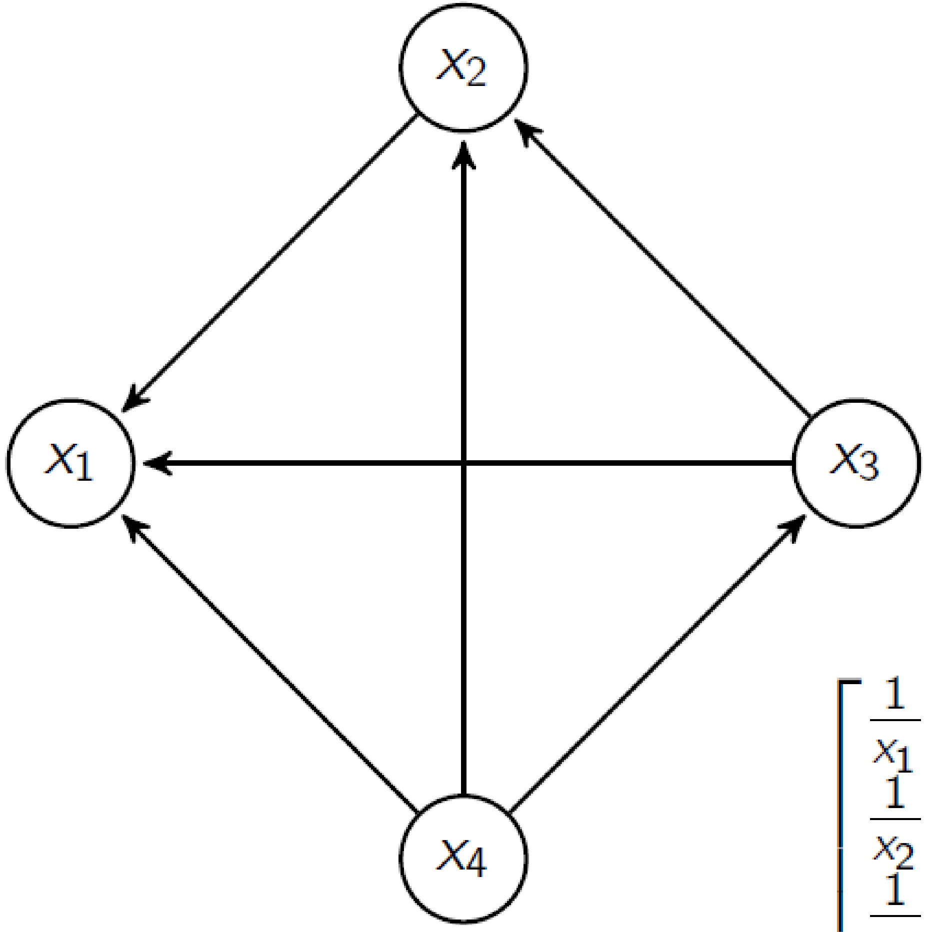
$$\frac{dy}{dt} = -y + \varepsilon\, xy$$

$\varepsilon > 0$ very small





Lotka-Volterra in 4D



Reference: H. Christodoulidi, A.N.W. Hone, T.E. Kouloukas, *A new class of integrable Lotka-Volterra systems*, Journal of Computational Dynamics, (2019).

$$\begin{bmatrix} \frac{1}{x_1}\frac{dx_1}{dt} \\ \frac{1}{x_2}\frac{dx_2}{dt} \\ \frac{1}{x_3}\frac{dx_3}{dt} \\ \frac{1}{x_4}\frac{dx_4}{dt} \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \end{bmatrix} + \underbrace{\begin{bmatrix} 0 & -\alpha_2 & -\alpha_3 & -\alpha_4 \\ \alpha_1 & 0 & -\alpha_3 & -\alpha_4 \\ \alpha_1 & \alpha_2 & 0 & -\alpha_4 \\ \alpha_1 & \alpha_2 & \alpha_3 & 0 \end{bmatrix}}_{Community\quad Matrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

Lotka-Volterra in 4D

We study a parametric family of (generalized) Lotka–Volterra systems of the form

$$\dot{x}_i = x_i \left(\sum_{j=1}^{4} A_{ij} x_j + r_i \right) , \qquad i = 1, \ldots, 4 ,$$

where $A = (A_{ij})$ is the community matrix:

$$A = \begin{bmatrix} 0 & -\alpha_2 & -\alpha_3 & -\alpha_4 \\ \alpha_1 & 0 & -\alpha_3 & -\alpha_4 \\ \alpha_1 & \alpha_2 & 0 & -\alpha_4 \\ \alpha_1 & \alpha_2 & \alpha_3 & 0 \end{bmatrix}$$

Lotka-Volterra in 4D

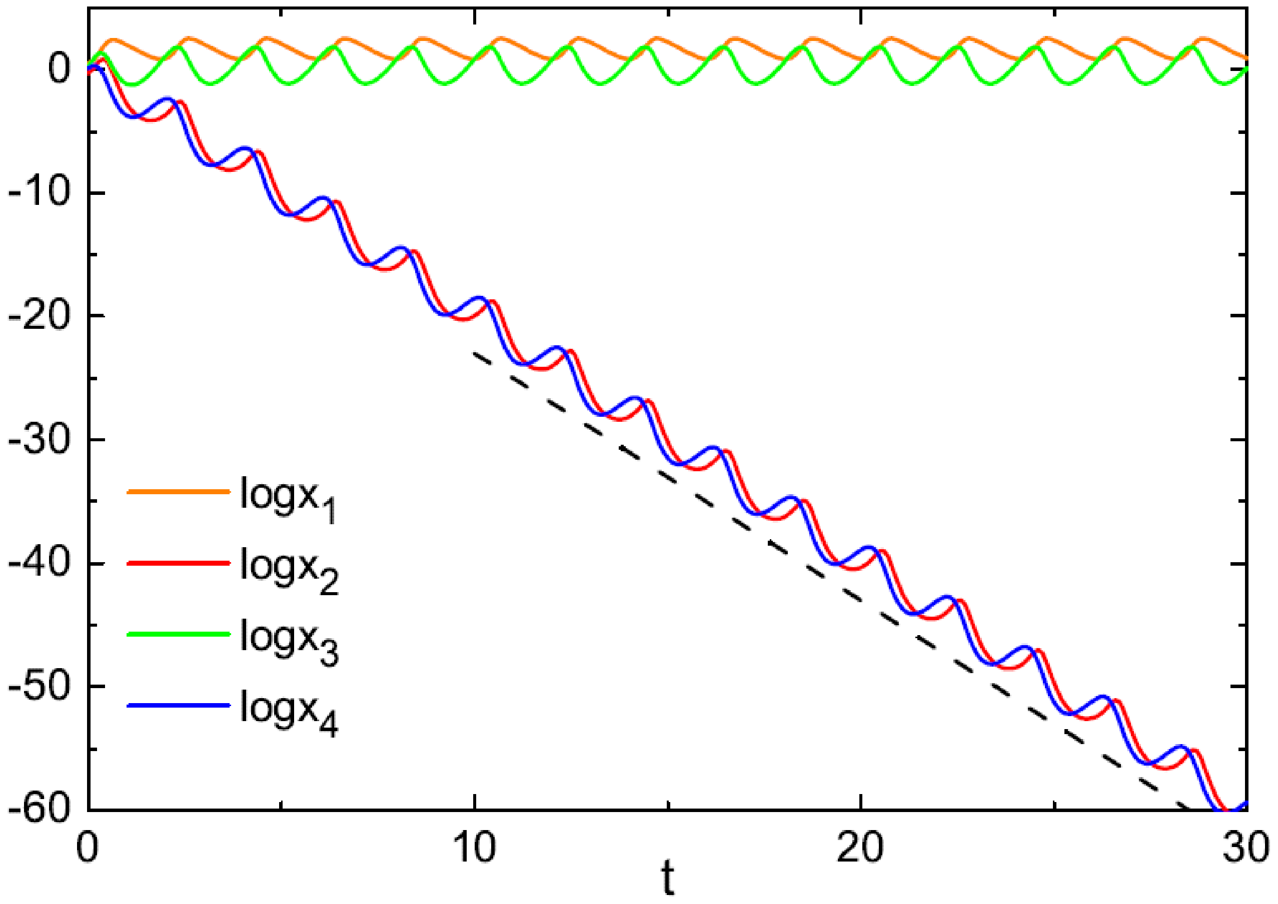


Figure: How the variables x_1, x_2, x_3, x_4 vary in time for $r = (-2, -2, -1, 1)$.

Lotka-Volterra in 4D

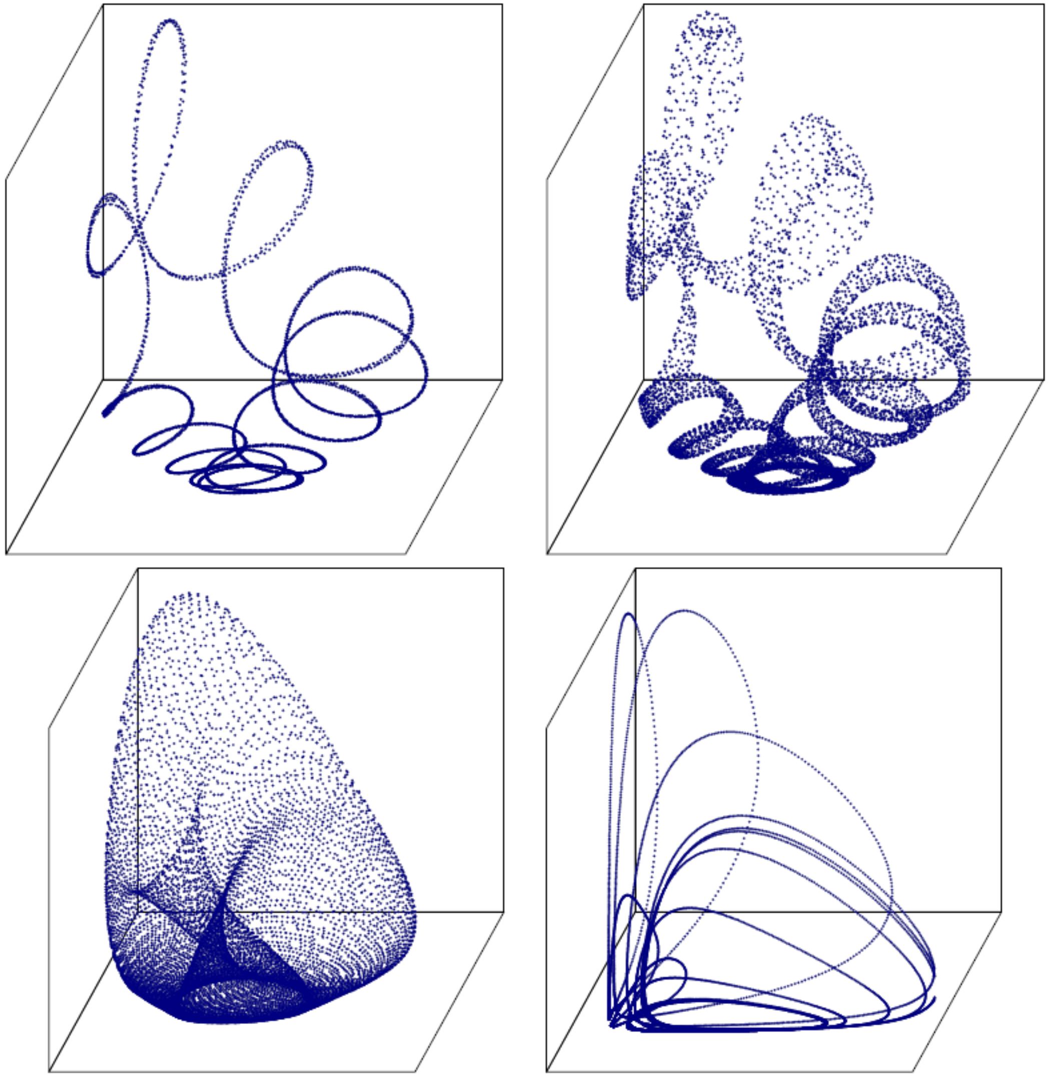


Figure: 3D projections on the x_2, x_3, x_4 plane for the Lotka–Volterra system with $(r_1, r_2, r_3, r_4) = (-3, -1, 1, 3)$. (a)–(c) for $E = 6$ (order) and (d) for $E = 20$ (chaos).

Lotka-Volterra in 4D

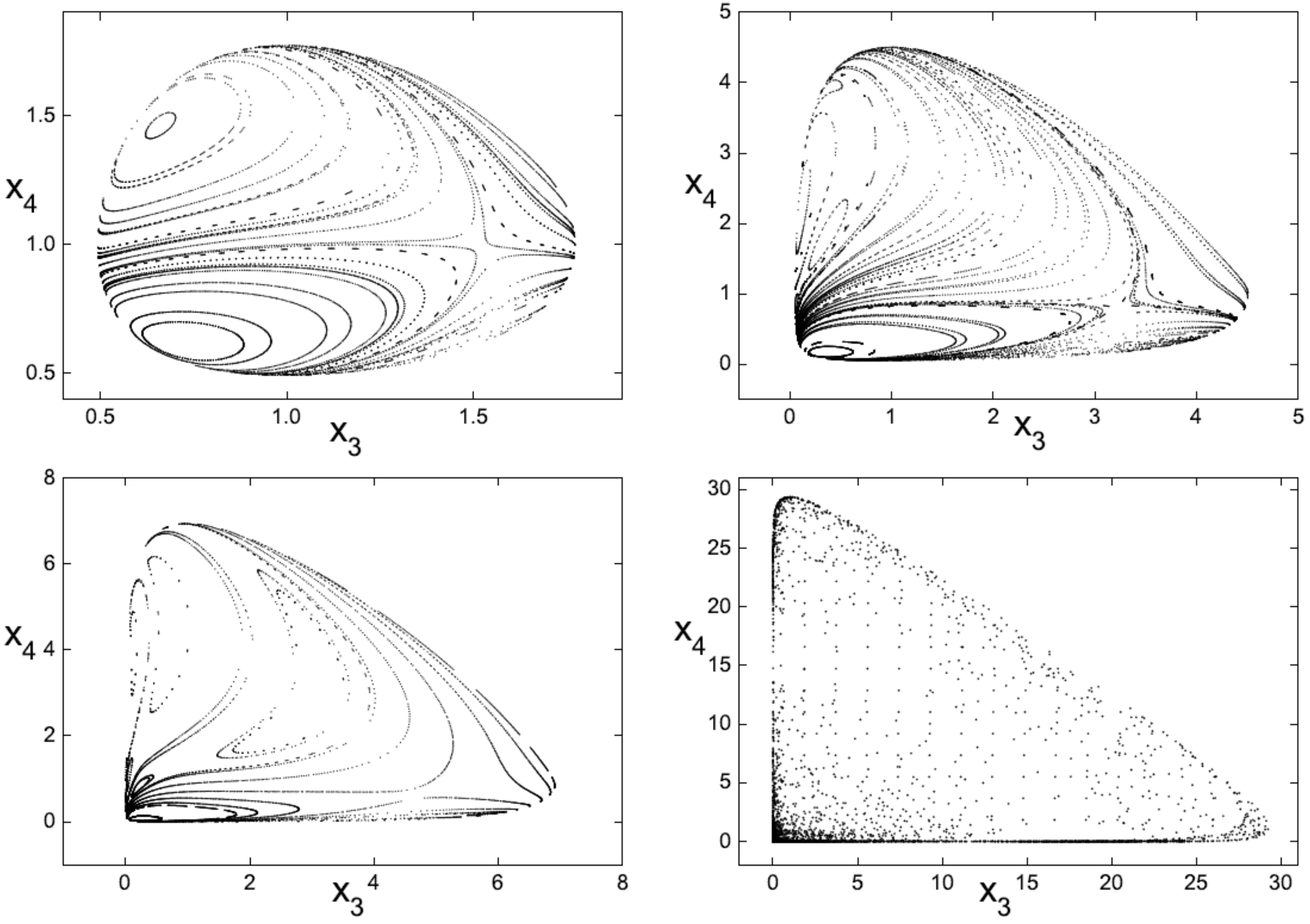


Figure: The Poincaré surface of section $x_2 = 1, x_1 > 1$ for the Lotka–Volterra system with $(r_1, r_2, r_3, r_4) = (-3, -1, 1, 3)$. Energies: (a) $E = 4.2$, (b) $E = 6$, (c) $E = 8$, (d) $E = 29$.

Conclusions

- The Lotka-Volterra system is an important predator-prey model which describes dynamical models in ecology, biology and finance.

- Case n = 2, the (original) Lotka-Volterra system, is solvable. There is no chaos. It is possible that both species survive.

- For n = 4 Lotka-Volterra systems there are cases with chaotic behaviour and cases where with organised (integrable) behaviour.
  - 4D chaotic cases are bounded, so all species co-exist at all times and display rich complexity.
  - 4D organised cases are unbounded: not all species survive.